

Joseph Flammia

5/5/2025

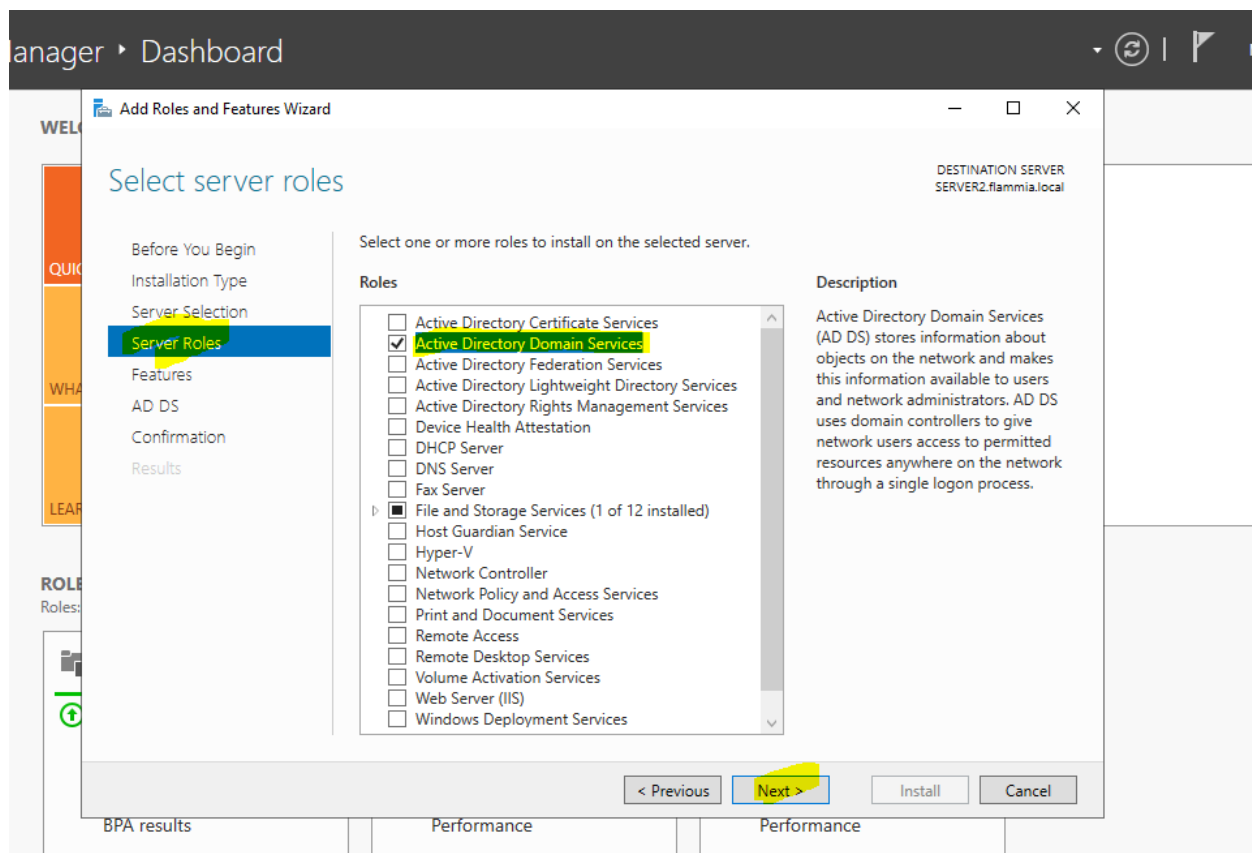
CYB126

Lab 26

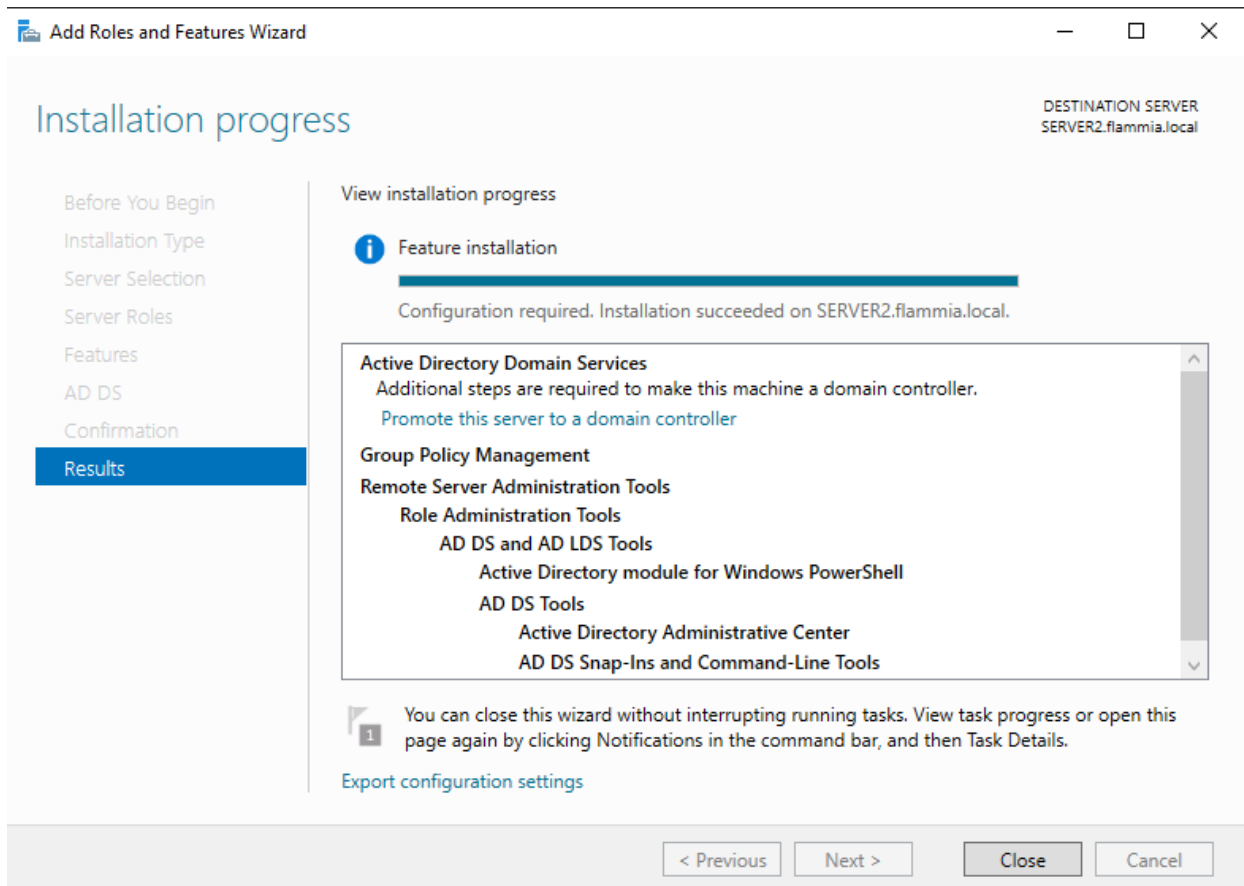
Lab 26A:

In this lab, we will be making DC2 of our Domain a DC. To start off, we will need to install AD on DC2.

To do this, simply open the Add Roles and Features Wizard, check off “Active Directory Domain Services”, click next, and then go through the features and “AD DS” tab.

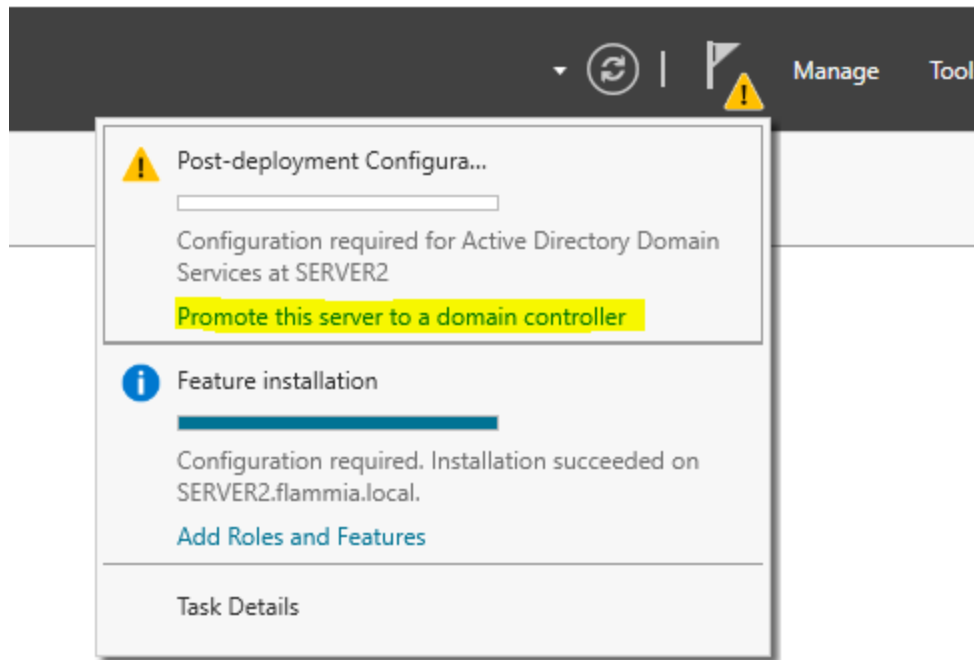


If everything checks out, we can begin the installation of AD on DC2. However, after installation, further configuration is required. We will need to promote DC2 to a domain controller.

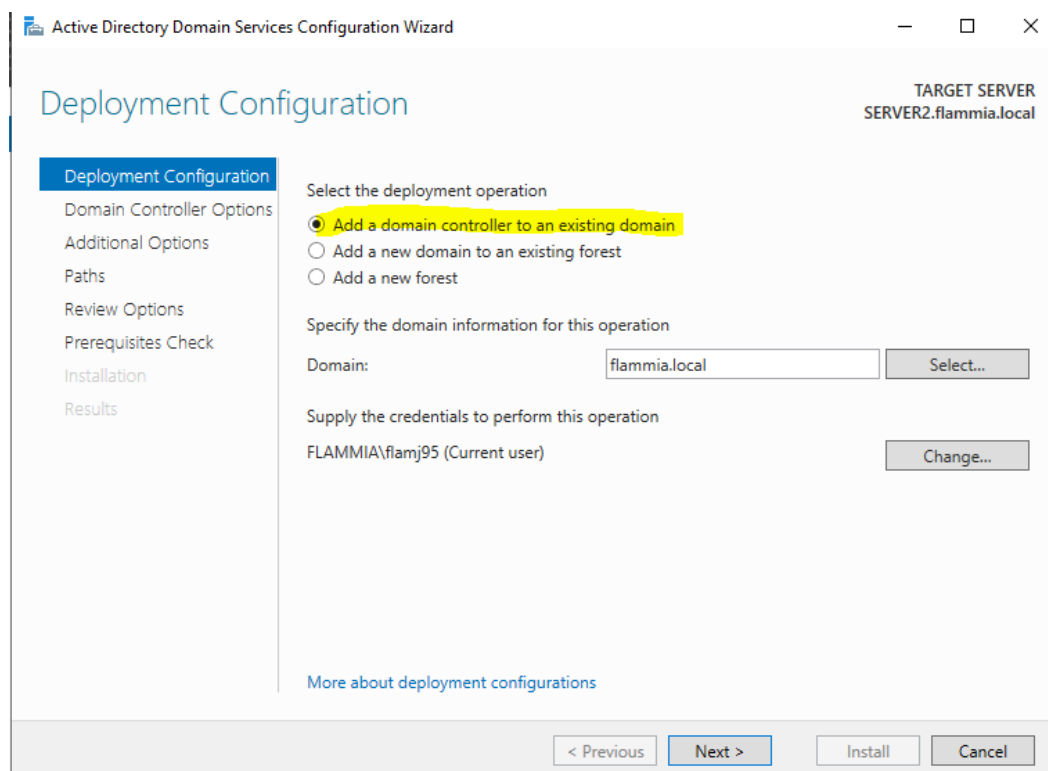


After installing AD, close the installation wizard and head back to the Server Dashboard. From here, click on the flag with the warning symbol on the top right.

As pictured below, we need to promote the server to a DC. All we need to do from here is click on “Promote this server to a domain controller”.



Clicking this will open the AD DS configuration wizard. On the “Deployment Configuration” tab, the only thing we will need to make sure is valid is the “Add a domain controller to an existing domain”. The rest of the options can be left as the default for this tab.

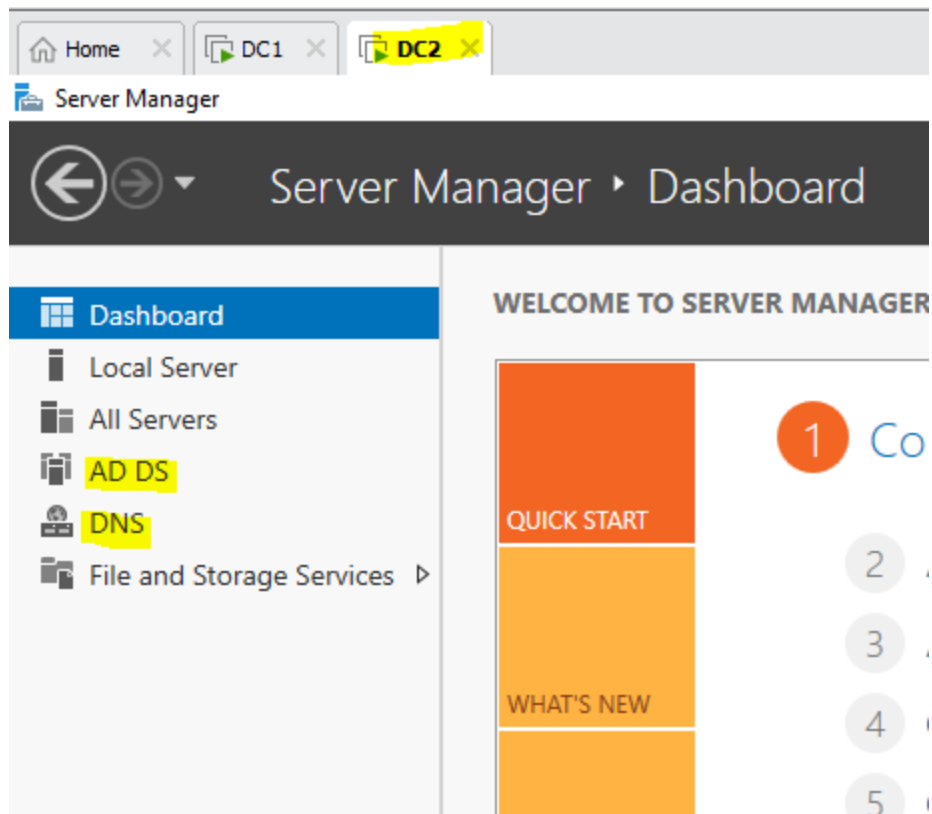


In the next tab, all we need to do here is provide a password just in case AD needs to go through maintenance or recovery tasks. The rest of the options on this tab can be left as the default.

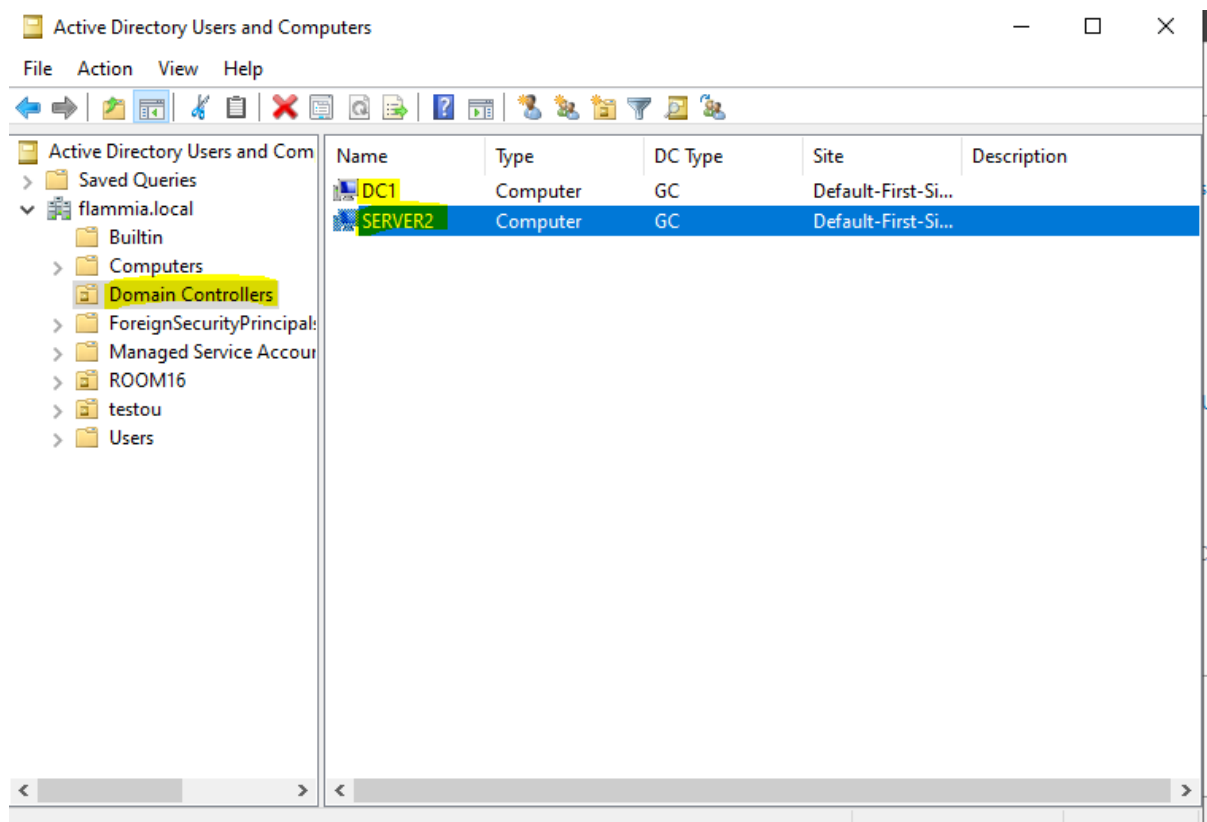
The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window. The title bar includes the window name and standard minimize, maximize, and close buttons. The main heading is 'Domain Controller Options'. In the top right corner, it identifies the 'TARGET SERVER' as 'SERVER2.flammia.local'. On the left, a vertical navigation pane lists the wizard's steps: 'Deployment Configuration', 'Domain Controller Options' (which is highlighted with a blue bar), 'DNS Options', 'Additional Options', 'Paths', 'Review Options', 'Prerequisites Check', 'Installation', and 'Results'. The main content area is titled 'Specify domain controller capabilities and site information'. It contains three checked checkboxes: 'Domain Name System (DNS) server', 'Global Catalog (GC)', and 'Read only domain controller (RODC)' (which is currently unchecked). Below these is a 'Site name' dropdown menu set to 'Default-First-Site-Name'. Further down, there is a section for the 'Directory Services Restore Mode (DSRM) password', with fields for 'Password:' and 'Confirm password:', both masked with dots. A blue link 'More about domain controller options' is located at the bottom of the main content area. The bottom of the window features a navigation bar with four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'.

The rest of the tabs' options can be left as the default. In the prerequisites check tab, just ensure everything looks correct and then begin the installation.

After installation, we will need to restart the server to confirm our changes. As you can see below, AD DS has been successfully installed.

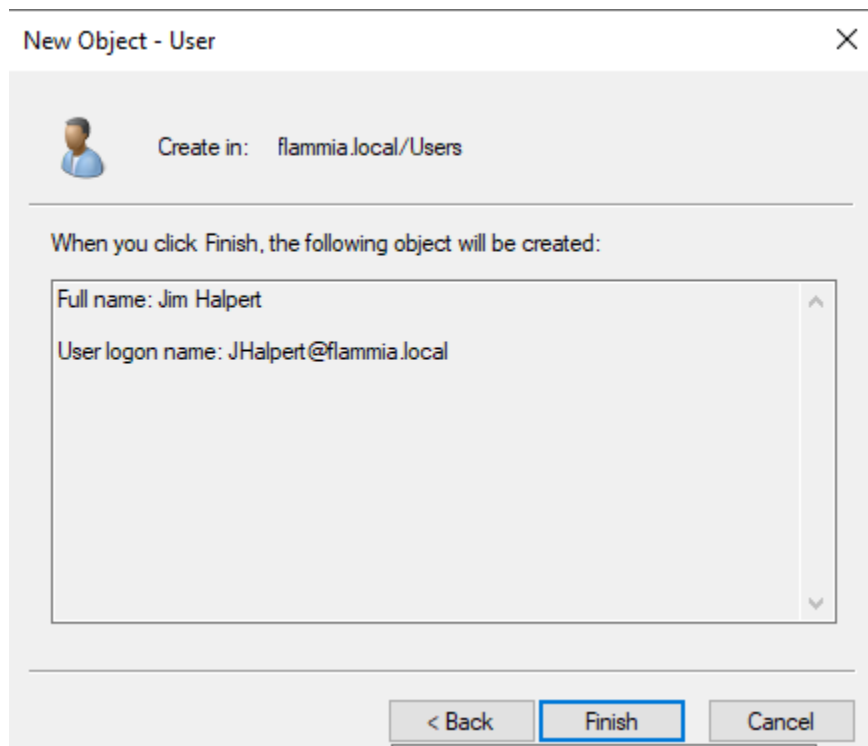


Just to confirm we have added DC2 as a Domain Controller, we can open the “Users and Computers” Application and check our current DCs. As you can see below, DC1 and SERVER2 (aka DC2) are both Domain Controllers.



To further test our configurations, we can try to make a user and group on one DC, and check if the user and group is present on the other.

As pictured below, I created a user called “JHalpert”, and a group called “QA” on DC2.



New Object - User

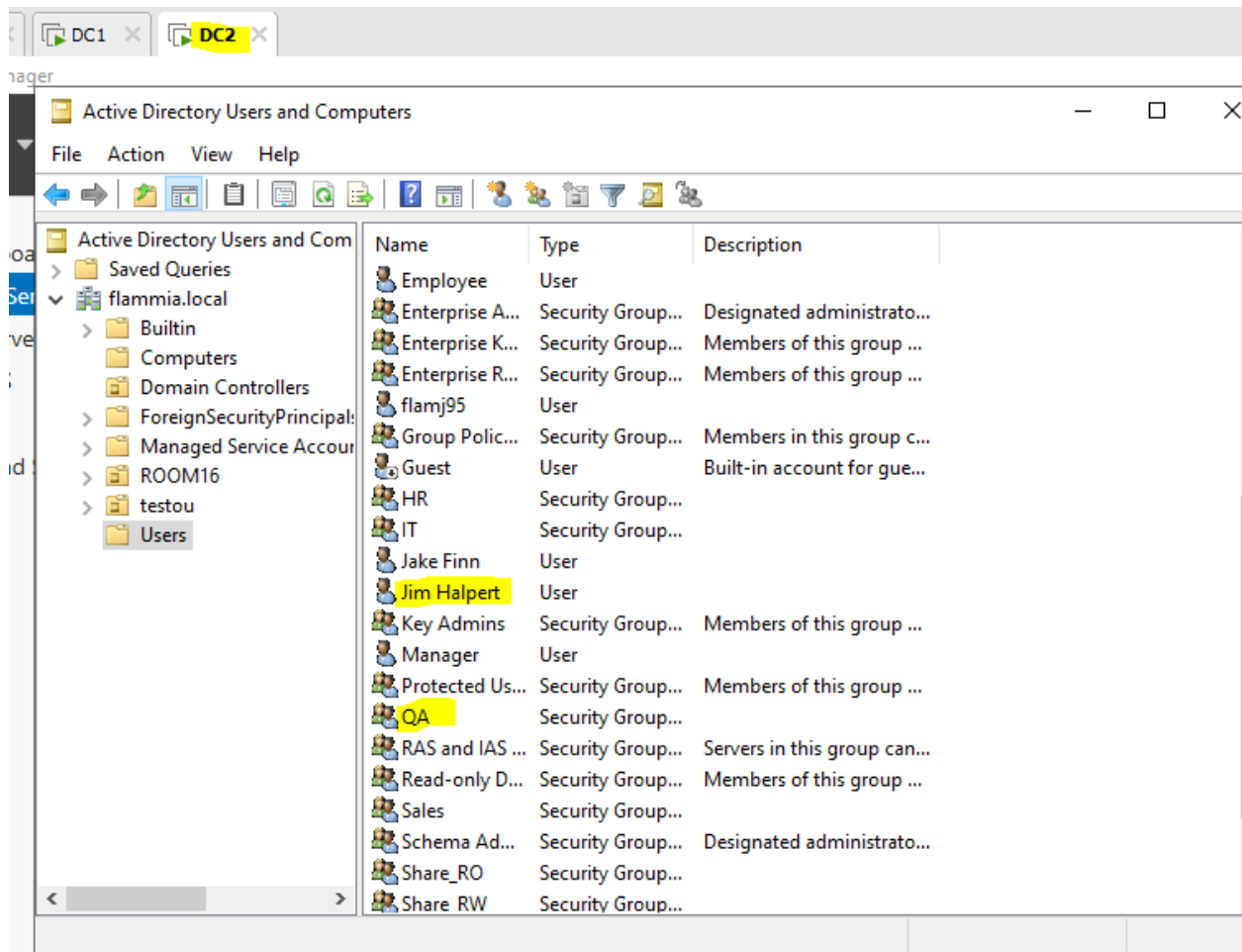
Create in: flammia.local/Users

When you click Finish, the following object will be created:

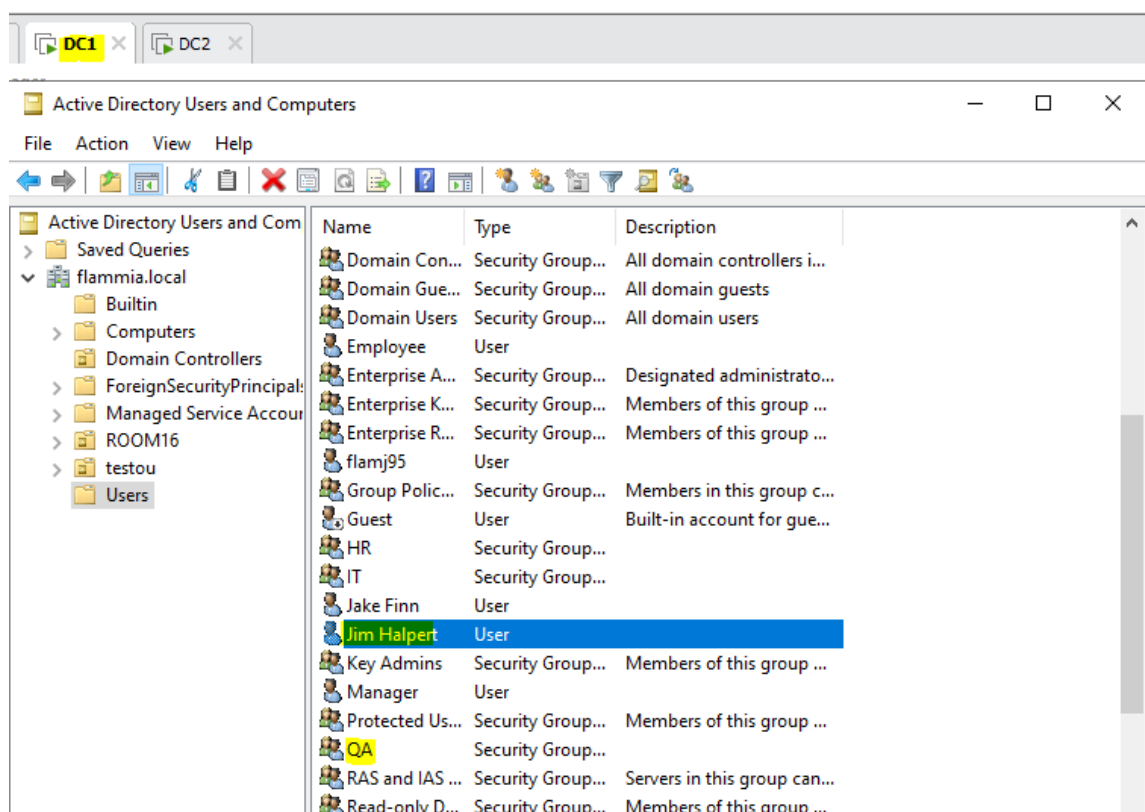
Full name: Jim Halpert

User logon name: JHalpert@flammia.local

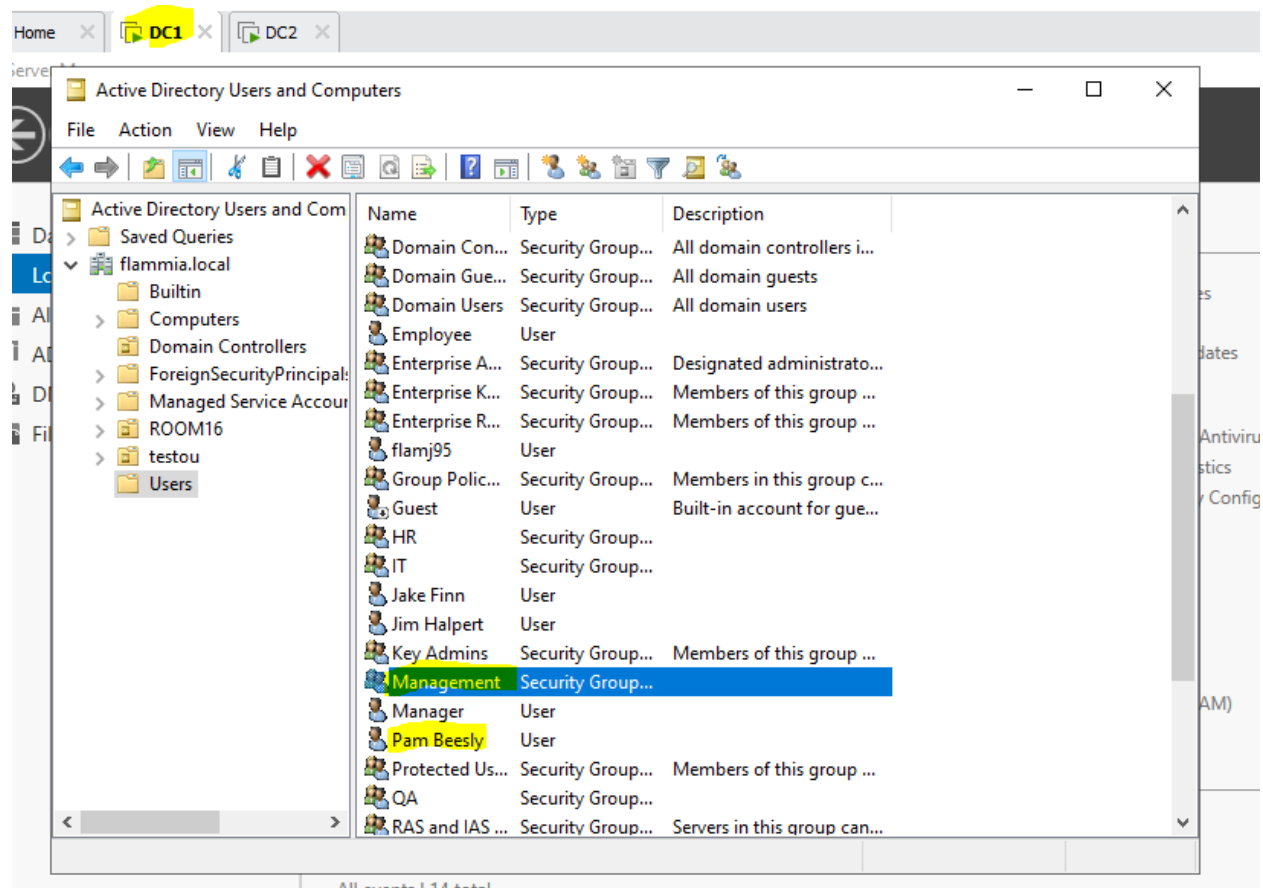
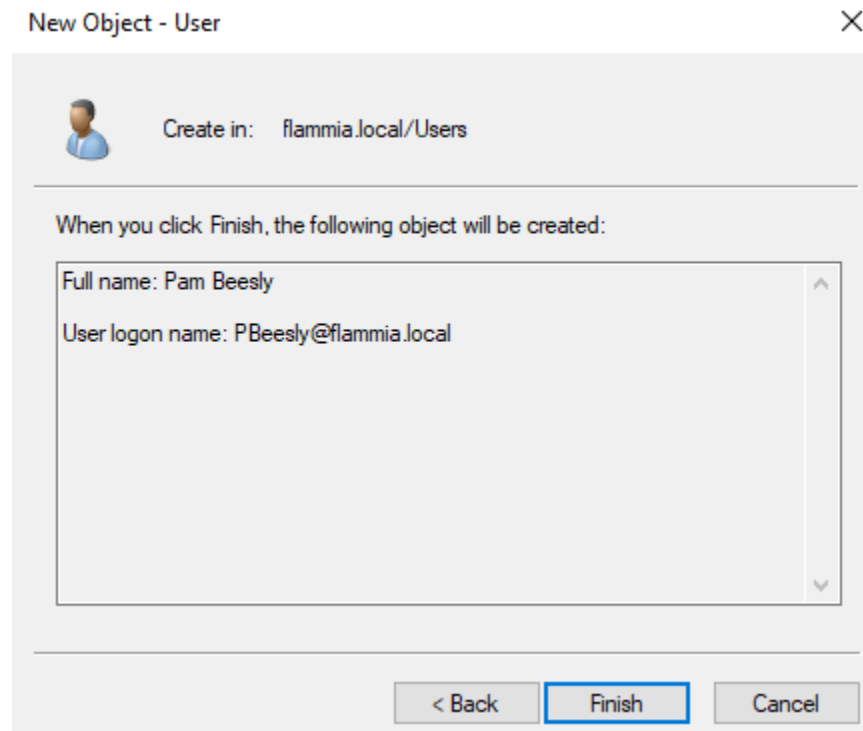
< Back Finish Cancel



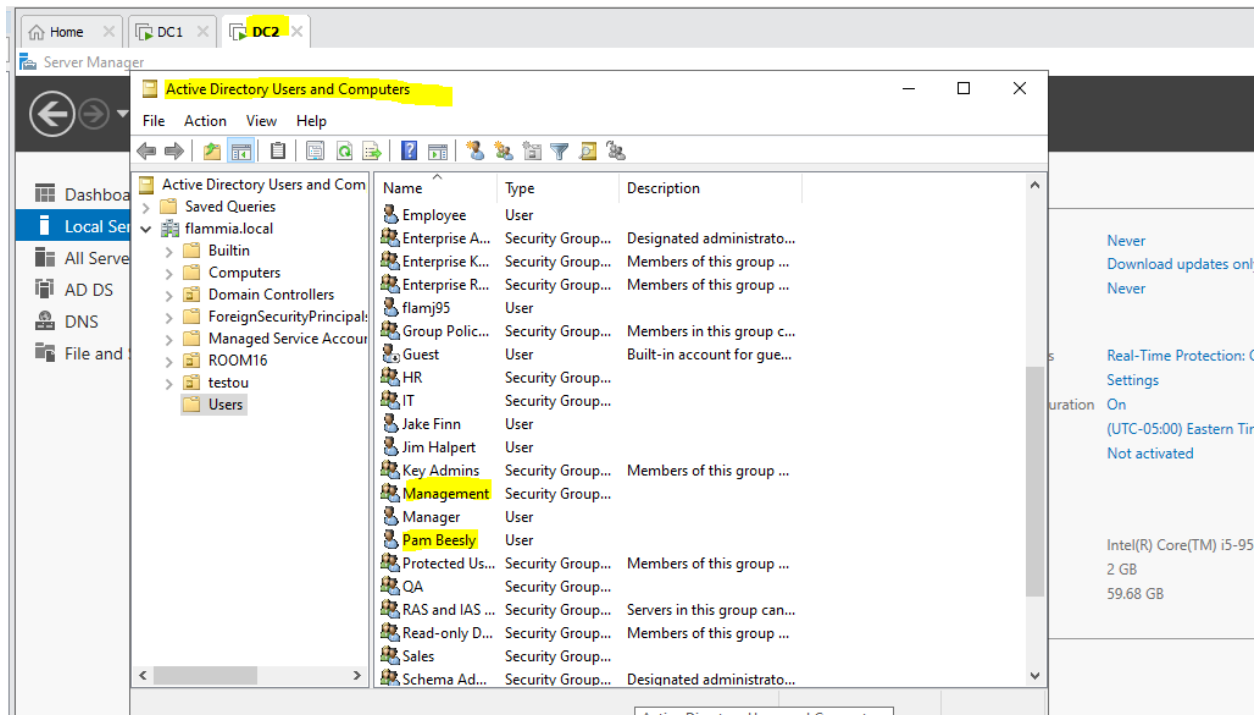
Switching back to DC1 and opening the Users and Computers application, we can see the newly created account and group are both present.



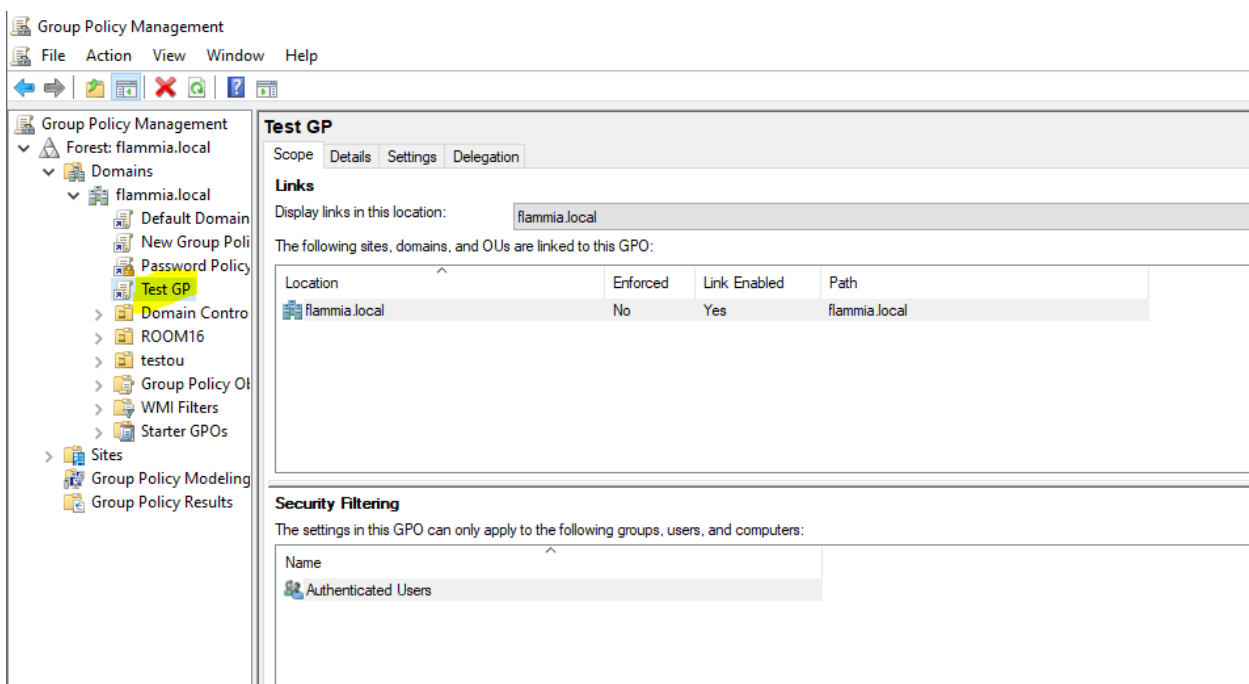
Now, let's try and do the same, except this time on DC1. As pictured below, I created an account named "PBeesly" and a group called "Management" on DC1.

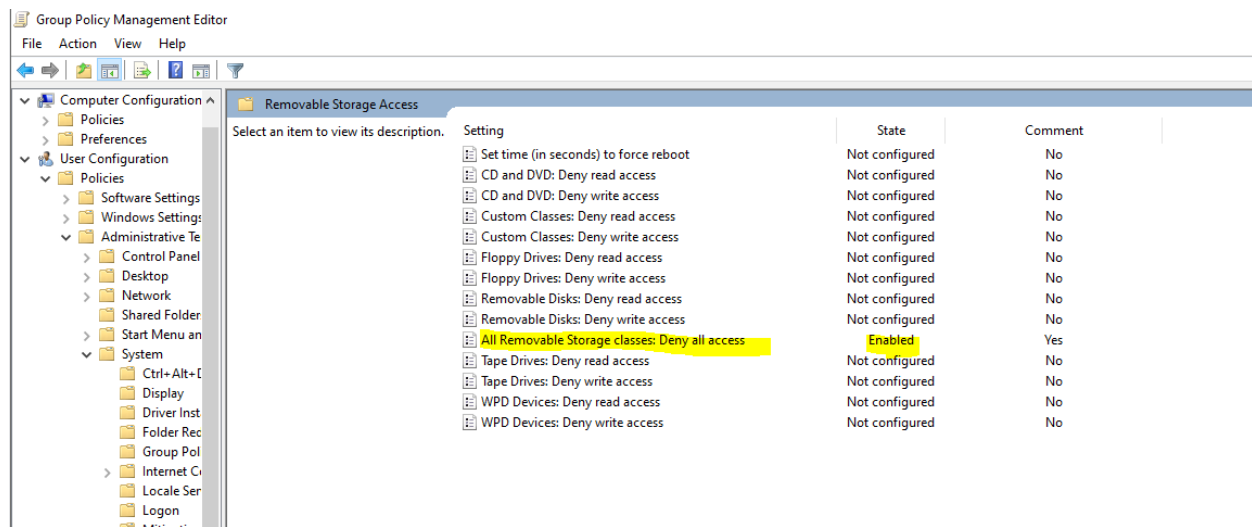


Now switching back to DC2, we can see that user and group are also present in the Users and Computers application.

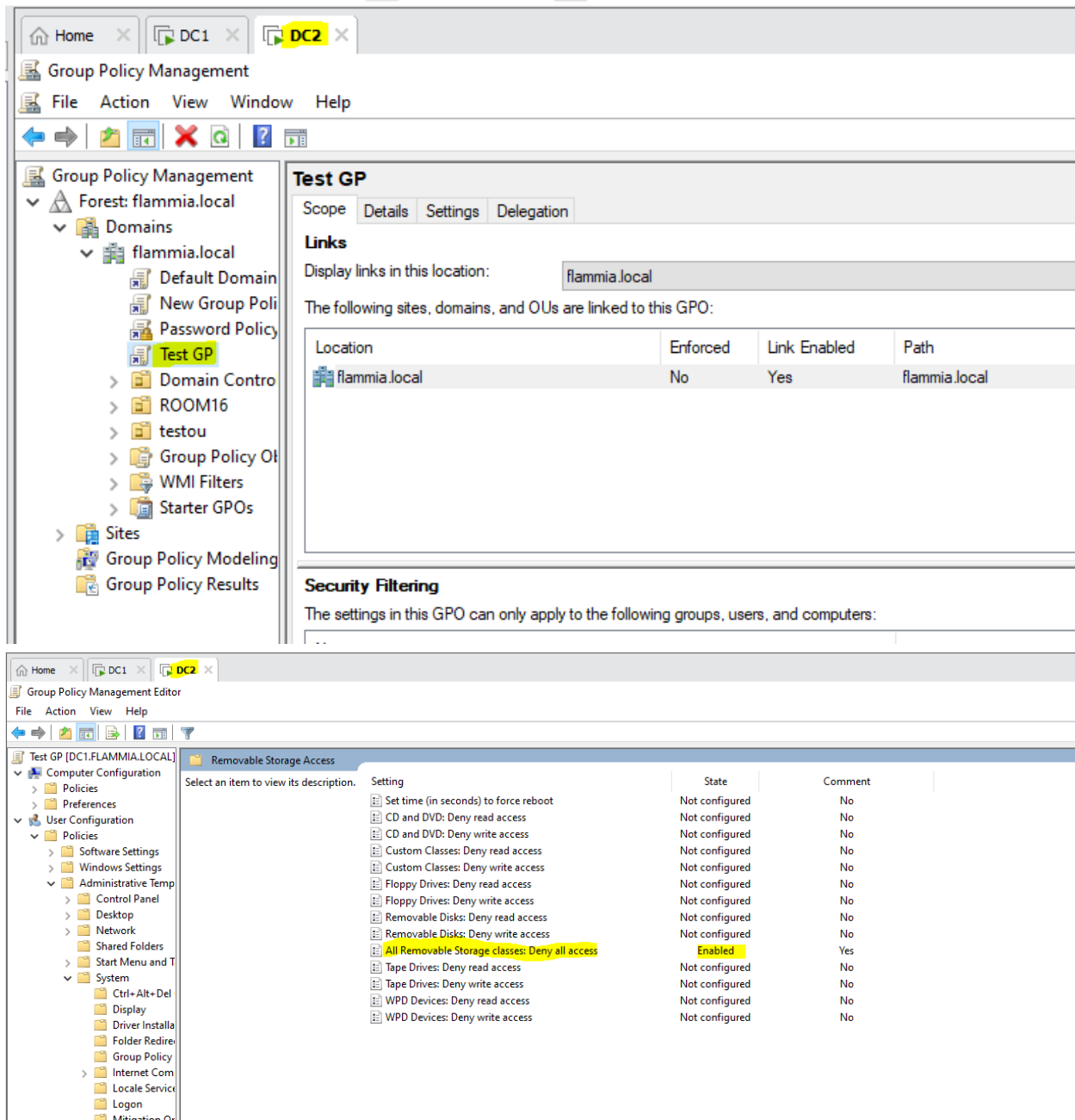


Next up, we will create a new group policy on DC1 and see if that transfers over to DC2. As pictured below, I created a new group policy called “Test GP” that blocks access to external storage devices (i.e USB drives)

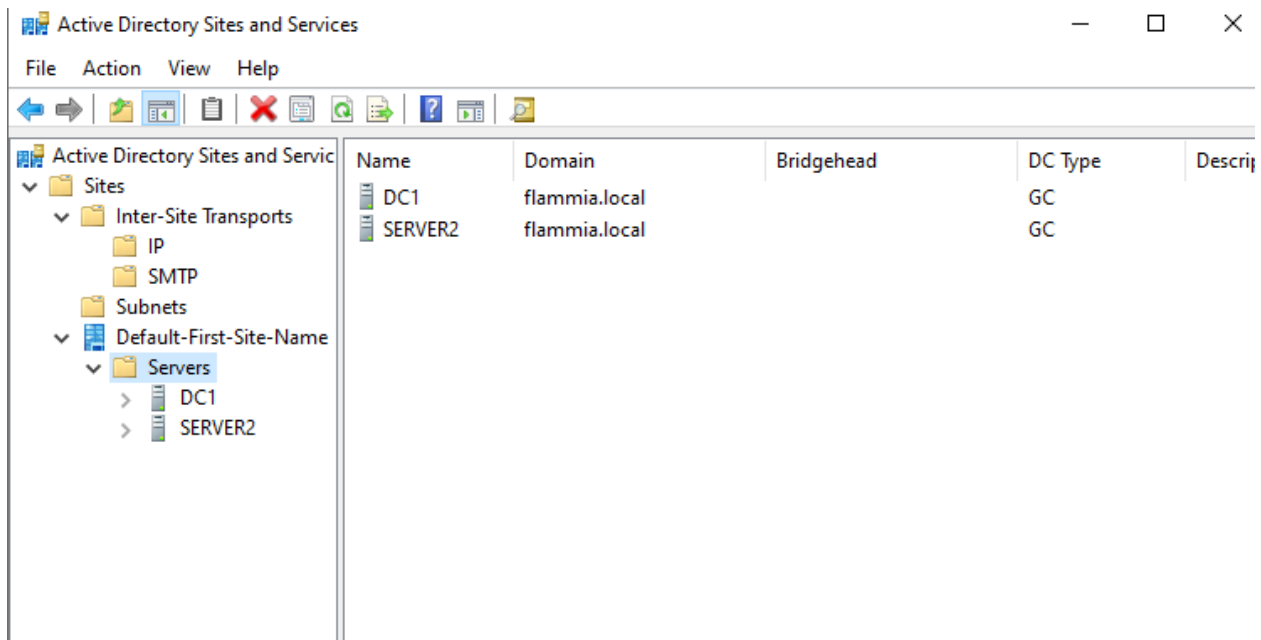




Now switching over to DC2 and opening the Group Policy application, we can see the “Test GP” policy has been successfully added.



Next, we will open up Active Directory Sites and Services. This application allows us to control replication data between domain controllers by defining sites and even defining subnets. As you can see below, by opening the “Servers” folder, our two Domain Controllers are present.



If we wanted to check the status of our domain controllers, we could do so with the command “repadmin /showrepl”. As pictured below, entering the command will output lots of information. This command essentially shows a report of your domain controllers and whether or not they are active and running. As we can see, SERVER2 (DC2) is running successfully as indicated by the “Last Attempt” line.

If there was an issue or an error with DC2, we could use this command to check and pinpoint the exact problem.

```

Administrator: Command Prompt
C:\Windows\system32>repadmin /showrepl

Repadmin: running command /showrepl against full DC localhost
Default-First-Site-Name\DC1
DSA Options: IS_GC
Site Options: (none)
DSA object GUID: dd53ce9c-0038-4141-962c-1a04ae84b3fd
DSA invocationID: caa799aa-1472-4a9e-b29d-09e82864ce7a

==== INBOUND NEIGHBORS =====
DC=flammia,DC=local
  Default-First-Site-Name\SERVER2 via RPC
    DSA object GUID: 89f30025-49ae-435d-bf2e-5716dfa11cee
    Last attempt @ 2025-05-05 14:14:36 was successful.

CN=Configuration,DC=flammia,DC=local
  Default-First-Site-Name\SERVER2 via RPC
    DSA object GUID: 89f30025-49ae-435d-bf2e-5716dfa11cee
    Last attempt @ 2025-05-05 13:58:00 was successful.

CN=Schema,CN=Configuration,DC=flammia,DC=local
  Default-First-Site-Name\SERVER2 via RPC
    DSA object GUID: 89f30025-49ae-435d-bf2e-5716dfa11cee
    Last attempt @ 2025-05-05 13:56:40 was successful.
  
```

A similar, more specific command to troubleshooting domain controllers is “repadmin /showsummary”.

This command will show exactly the replication status of the Domain Controllers, and if there are any errors present.

```
C:\Windows\system32>repadmin /replsummary
Replication Summary Start Time: 2025-05-05 14:29:27

Beginning data collection for replication summary, this may take awhile:
.....

Source DSA          largest delta      fails/total %%      error
DC1                  36m:11s           0 / 5    0
SERVER2              32m:38s           0 / 5    0

Destination DSA     largest delta      fails/total %%      error
DC1                  32m:38s           0 / 5    0
SERVER2              36m:11s           0 / 5    0
```

We can check FSMO roles by entering “netdom query fsmo” into the command prompt. As you can see, all roles are currently assigned towards DC1. Each role has an integral part to the domain.

Schema Master:

The Schema Master is responsible for updating any type of attributes, roles, or objects within the domain. For example, by creating a new user on DC2, the Schema Master updates the schema, so the change is reflected across the domain.

Domain Naming Master:

As the name suggests, the Domain Naming Master is responsible for maintaining domain records and domain name uniqueness across the forest.

PDC: The PDC emulator in a domain is responsible for time synchronization and password information across the domain. For example, by changing the password on a client in the domain that points towards DC1, you can login to a different PC that points towards DC2. DC2 obtains the new password information via the PDC.

RID Pool Manager:

The RID Pool Manager in a domain is responsible for simply ensuring each security feature (for example, a group or user) has a unique SID. For example, when creating a new security group on DC2, DC2 requests a new SID from DC1.

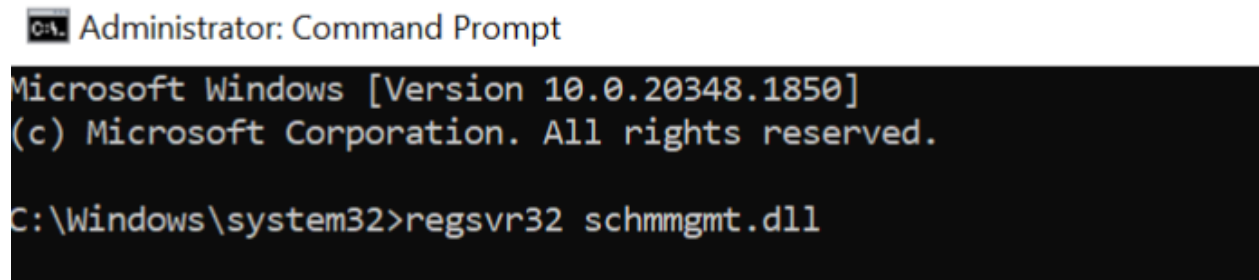
Infrastructure Master:

The Infrastructure Master in a domain is responsible for translating SID's, GUID'S, and DN'S between Domains. This role is especially important if you have multiple domains in your forest.

```
C:\Windows\system32>netdom query fsmo
Schema master           DC1.flammia.local
Domain naming master    DC1.flammia.local
PDC                     DC1.flammia.local
RID pool manager        DC1.flammia.local
Infrastructure master    DC1.flammia.local
The command completed successfully.
```

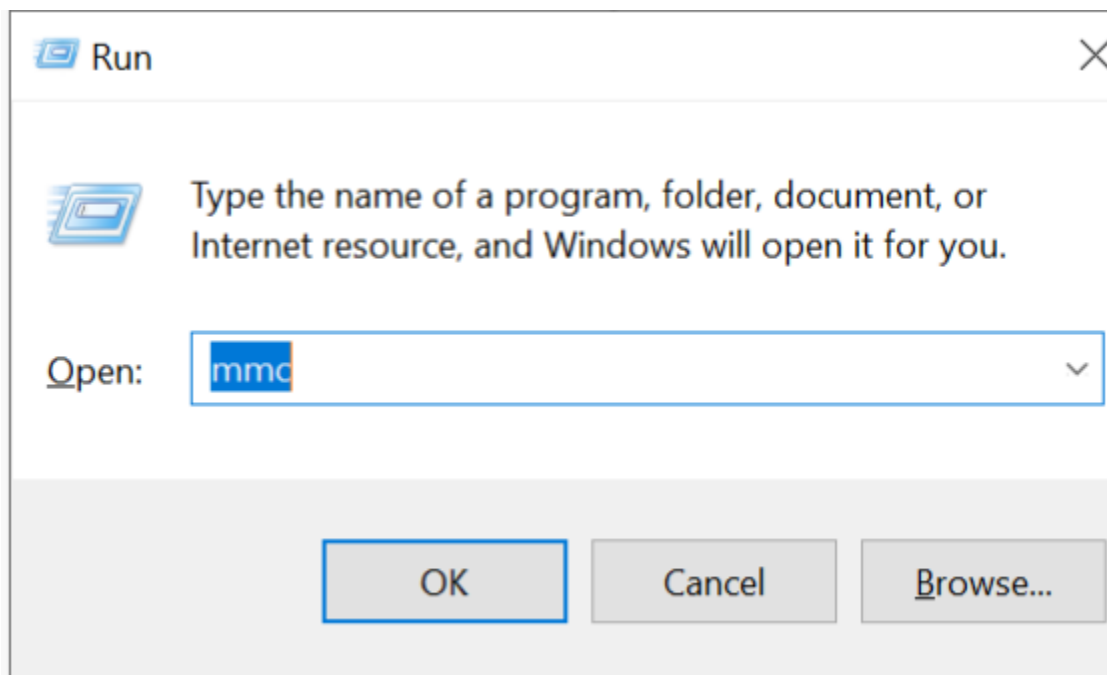
If we wanted to transfer FSMO roles between Domain Controllers, specifically the Schema Master and Domain Naming Master roles, we could do so opening the Users and Computers application.

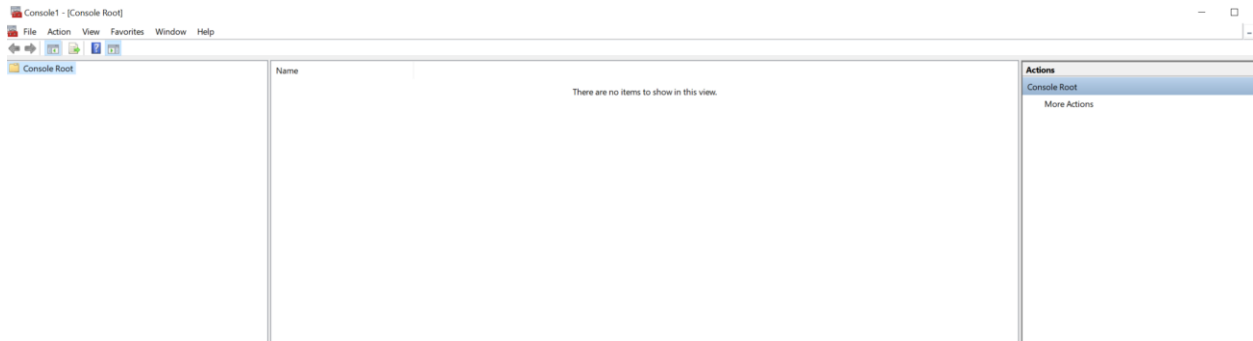
First, we will transfer the schema master role to DC2. To start, open an administrative command prompt on DC1 and enter the command “regsvr32 schmmgmt.dll”.



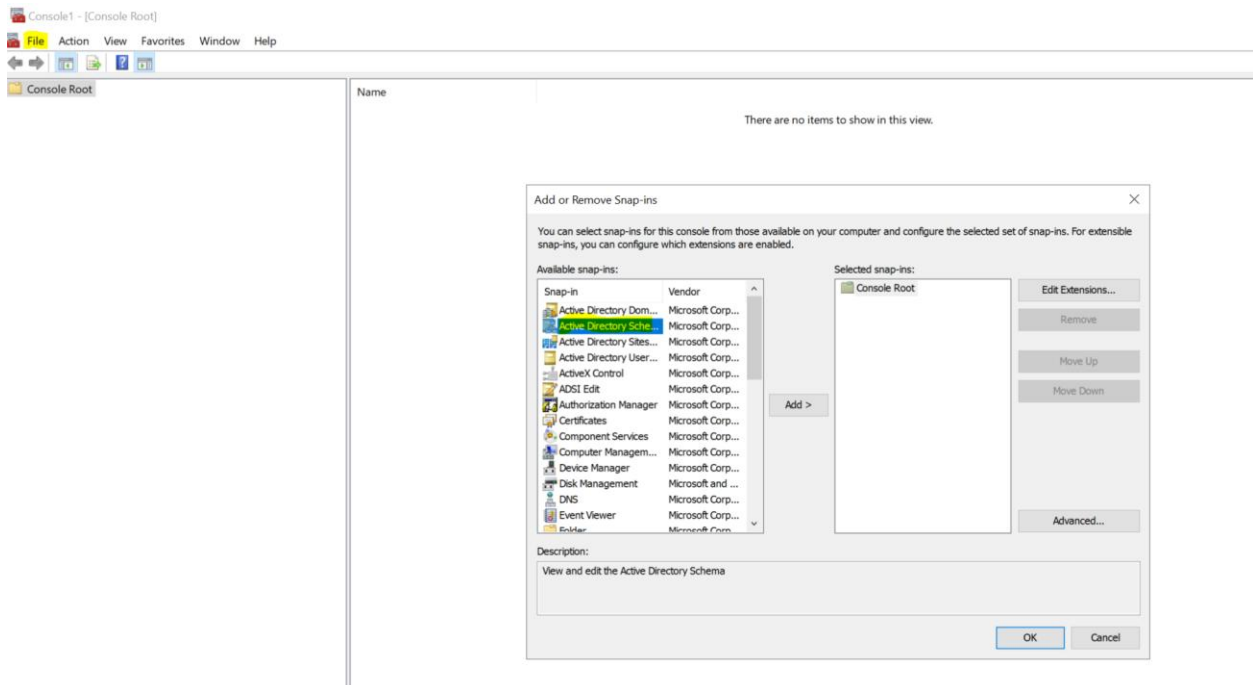
```
C:\> Administrator: Command Prompt  
Microsoft Windows [Version 10.0.20348.1850]  
(c) Microsoft Corporation. All rights reserved.  
C:\Windows\system32>regsvr32 schmmgmt.dll
```

Next, hold down the Windows key + R to open the run application, and enter “mmc” to open the Microsoft Management Console.

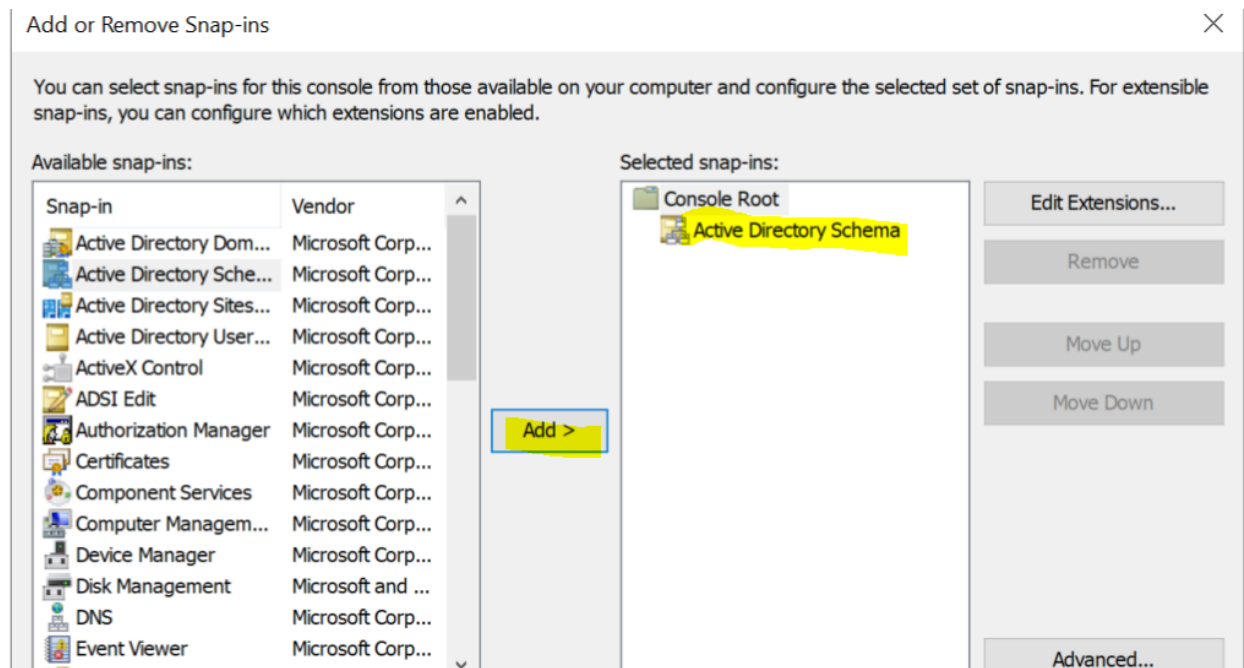




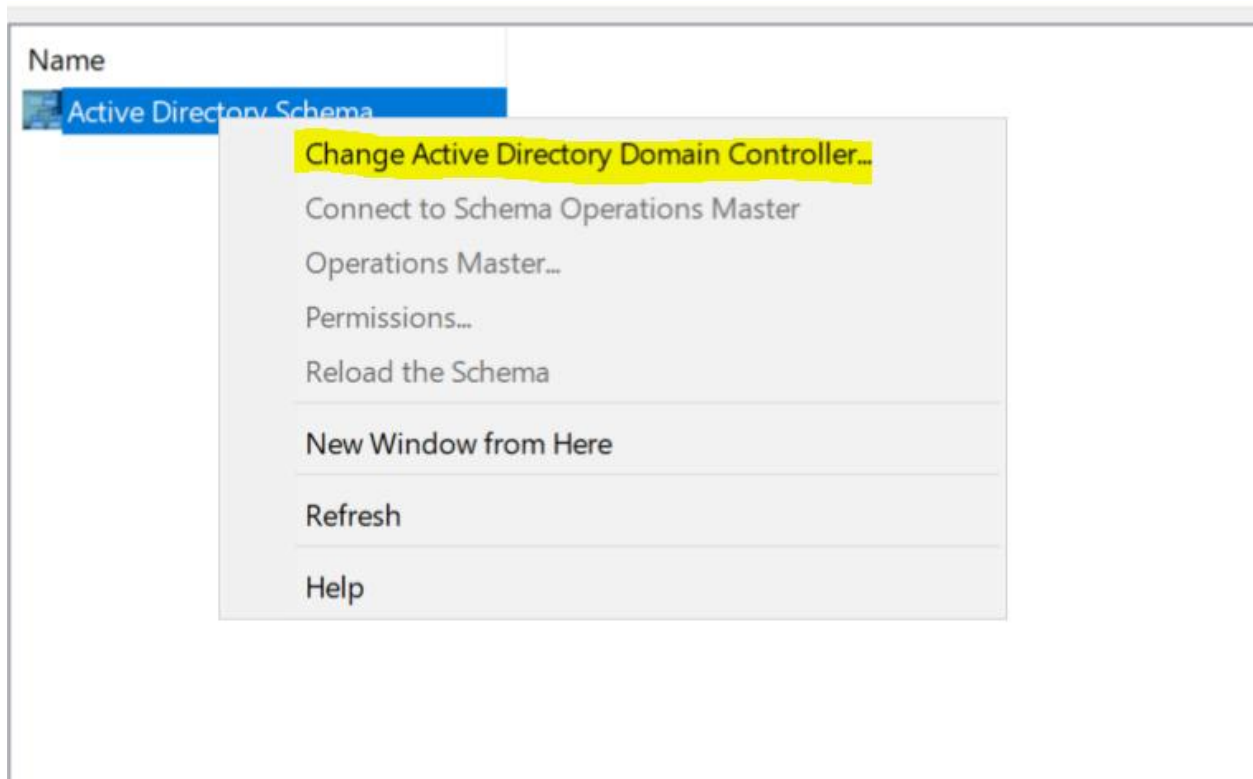
From here, click on "File", then "Add/Remove Snap in", and then find the "Active Directory Schema" option.



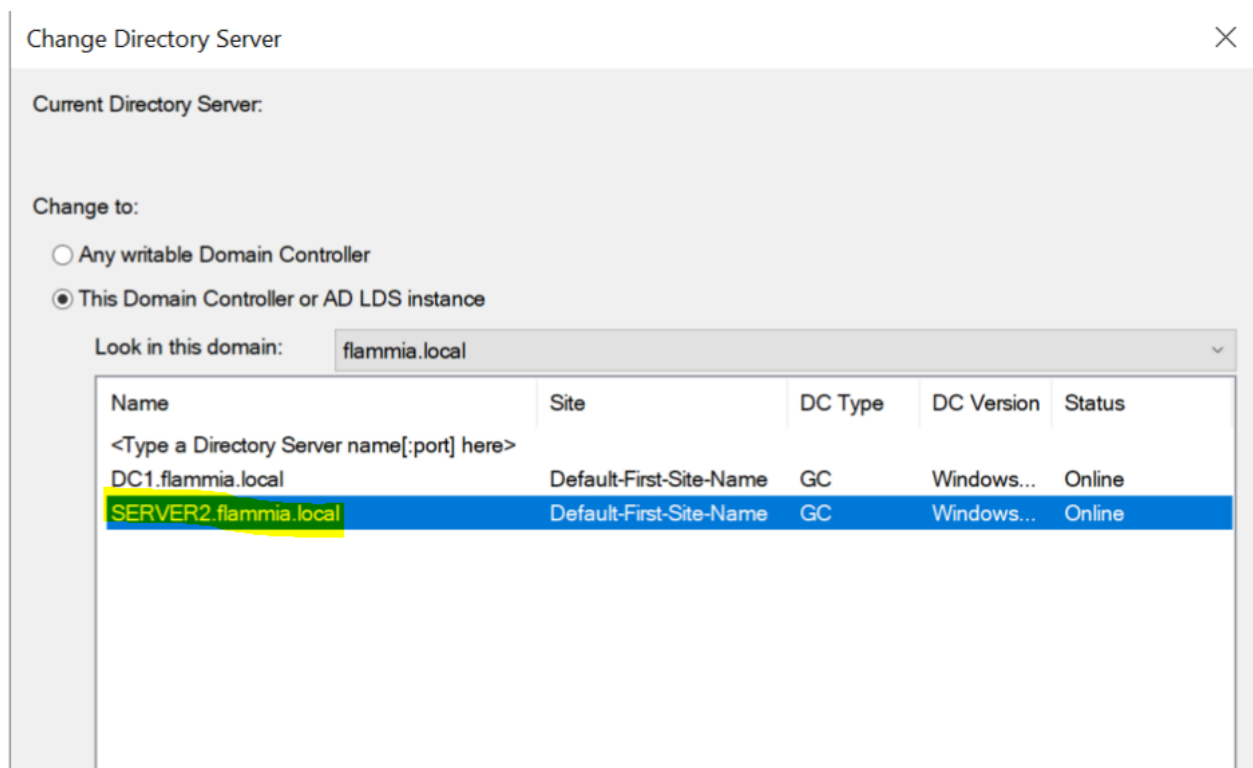
Next, click on "Add" to add the feature to the management console, and then click "OK" to save our changes.



From here, right click on the “Schema” option that was now added to the console, and click “Change Active Directory Domain Controller”.

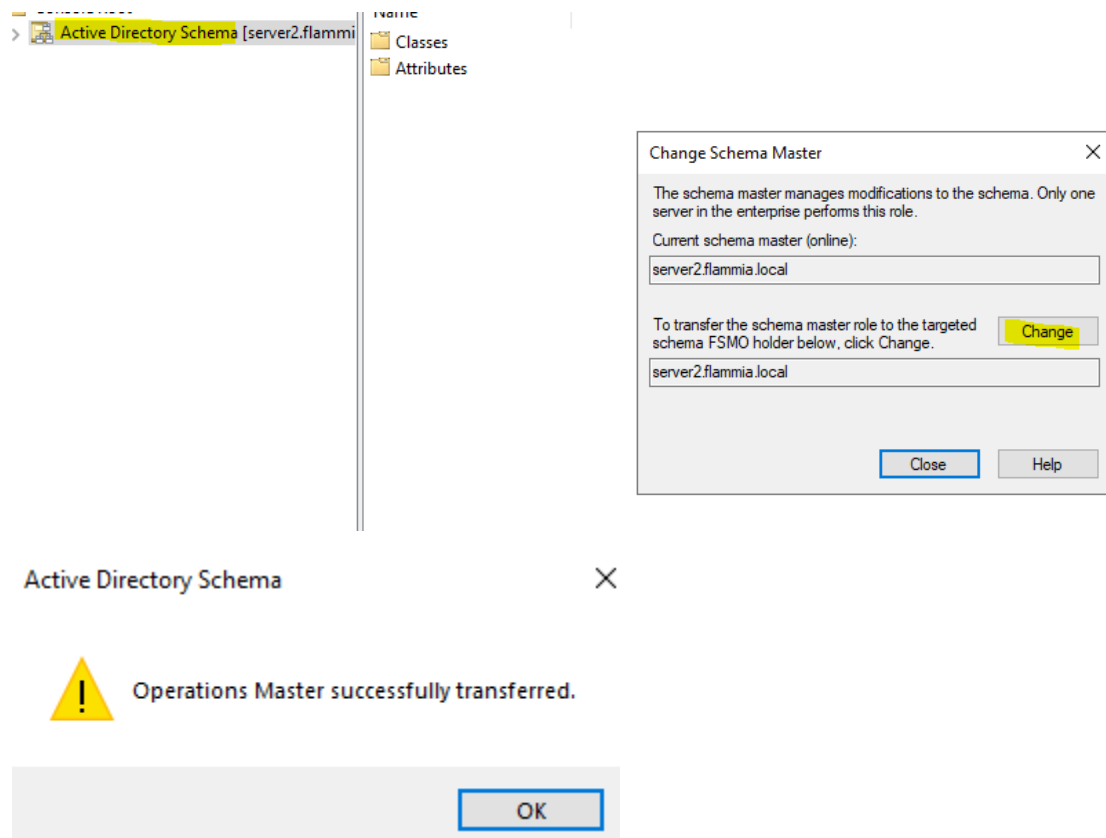


All we have left are two things to do, first off select the Domain Controller we want to transfer the Schema Master role to. In my case, I am choosing DC2 (SERVER2). Once done, simply click “OK”.



Next, we will need to confirm our changes. To do this, simply right click on the Schema tab, click “Operations Master” and then click the “change” button highlighted below.

NOTE: For me, the “Current Schema Master” is listed as “server2.flammia.local”. This is because I already changed the schema master. Normally, this would be the DNS record of the primary DC (In our case, DC1)



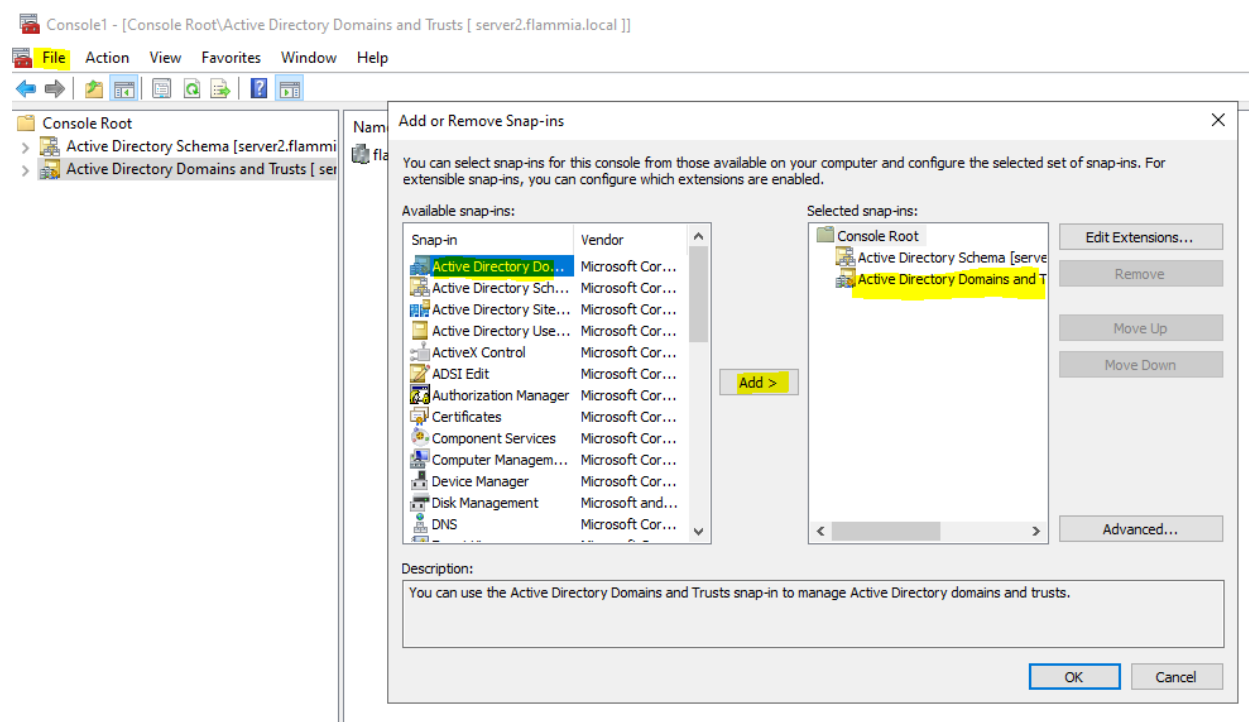
After transferring the ownership of the role, we can check if our configurations were successful by running the “netdom query fsmo” command.

As you can see, the schema master is now SERVER2.

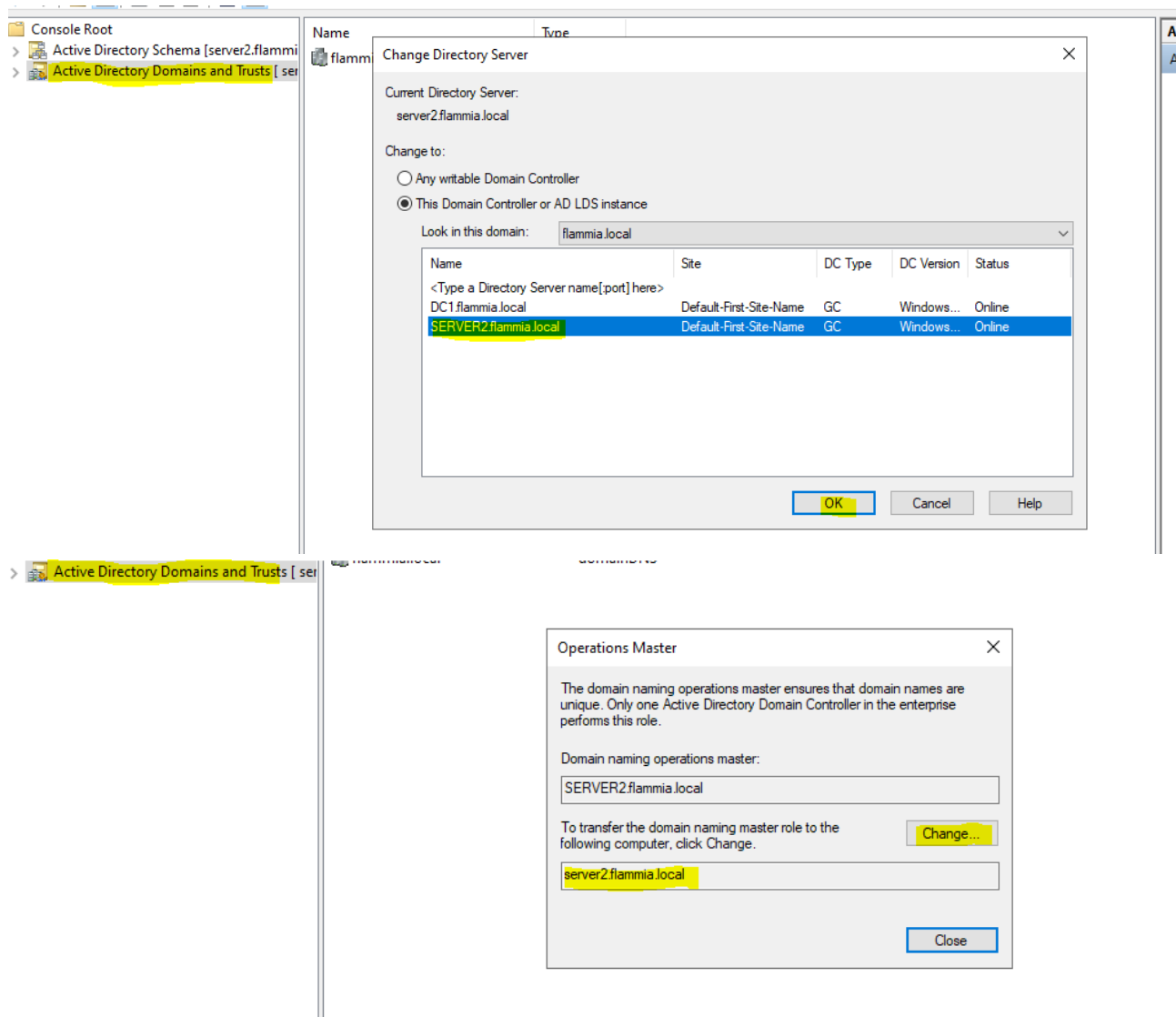
```
C:\Users\Administrator>netdom query fsmo
Schema master          SERVER2.flammia.local
Domain naming master   DC1.flammia.local
PDC                   DC1.flammia.local
RID pool manager       DC1.flammia.local
Infrastructure master  DC1.flammia.local
The command completed successfully.
```

Next, let's try to change the Domain Naming Master role to DC2. This process will be nearly identical as before. However, instead of choosing the "Schema" snap-in, we will choose the "Domains and Trusts" snap-in.

As pictured below, I opened MMC, clicked on "File", then "Add/Remove Snap-in". From here, I chose the "AD Domains and Trusts" option, clicked "Add", and then "OK" to save our changes.



Then, I changed the DC and Operations Master of the role to DC2 as shown below:



Again, just to confirm our role transfers were successful, I ran the “netdom query fsmo” command in a command prompt and got the following results:

```
C:\Users\Administrator>netdom query fsmo
Schema master          SERVER2.flammia.local
Domain naming master   SERVER2.flammia.local
PDC                    DC1.flammia.local
RID pool manager       DC1.flammia.local
Infrastructure master  DC1.flammia.local
The command completed successfully.
```